INCH-POUND

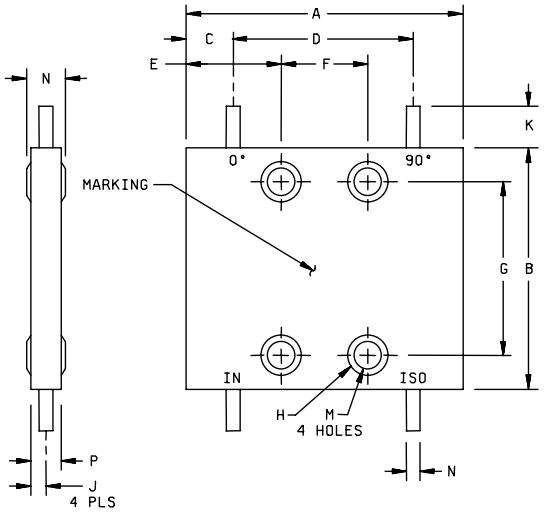
MIL-DTL-23971/12A
23 April 2002
SUPERSEDING
MIL-P-23971/12
22 February 1980

DETAIL SPECIFICATION SHEET

POWER DIVIDER/COMBINERS, QUADRATURE STRIPLINE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the power divider described herein shall consist of this specification sheet and MIL-DTL-23971.



	Dimensions			
	Inches		Millimeters	
	Max	Min	Max	Min
A	See Table I			
B				
C				
D				
E				
F				
G				
H				
J	.065	.045	1.65	1.14
K	N/A	.15	N/A	3.8
L	.06	.04	1.5	1.1
M	.105 dia	.093 dia	2.67 dia	2.36 dia
N	.14	N/A	3.56	N/A
P	.11	N/A	2.79	N/A

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Dimensions and config. dash nos. 01 and 02.

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TABLE I. Dimensions.

M23971 /12-	A	B	C	D	E	F	G	H
01	1.83 ±.02	2.02 ±.02	.18 ±.02	1.460 ±.015	.37 ±.02	1.060 ±.010	1.750 ±.015	.22 max
02	1.00 ±.01	.87 ±.01	.17 ±.01	.66 ±.01	.343 ±.010	.313 ±.010	.625 ±.010	N/A

TABLE II. Electrical performance characteristics and physical requirements.

M23971 /12-	Imped-ance (ohms)	Frequency range (GHz)	Average coupling (dB)	VSWR max	Insertion loss max (dB)	Isolation min (dB)	Phase balance max (degree)	Amplitude balance (dB)	Power level		Weight pounds (grams)	Ambient temperature		Figure
									avg (W)	pk (W)		Operating	Storage	
01	50	.1-.16	3. +.2, -.0	1.2:1	0.3	25	±2.0	±0.6	200	1K	.038 (18)	-55° to +71°C	-55° to +65°C	1
02	50	.5-1.0	3. +.2 -.0	1.2:1	0.25	20	±1.5	±0.5	200	1K	.054 (25)	-20° to +75°C	-55° to +85°C	1

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REQUIREMENTS:

Design and construction: See figure 1.

Housing: Hermetically sealed.

Material:
Dash numbers 01 and 02: 6061-T6 aluminum.

Finish:
Dash number 01: Clean iridite finish per MIL-C-5541, type 1, grade C class III.
Dash number 02: Gold plated per SAE-AMS-2422, .000050 to .000100 inch thick over 2-ounce copper clad microstrip.

Terminals:

Material:

Dash number 01: Nickel alloy.
Dash number 02: Silphos (15% silver, 80% copper, 5% phosphorus).

Finish:
Dash number 01: Tin plated.
Dash number 02: Nickel clad, .004 inch thick nickel plus .001 thick silphos. Solder dipped with Sn60 or Sn63 per American National Standards J-STD-004, J-STD-005 and J-STD-006.

Nickel is to be used only if specification cannot be met using alternate material means.

Electrical characteristics: See table II.

Weight: See table II.

Ambient temperature: See table II.

Environmental tests: In accordance with MIL-DTl-23971 except:
Dash number 02:
Vibration, high frequency: In accordance with method 204 of MIL-STD-202, test condition B.

Part number: M23971/12-(dash number from table II).

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Navy - AS, MC, OS

Preparing activity:
DLA - CC

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